

UNIVERSITY OF BIRMINGHAM

School of Mathematics

Programmes in the School of Mathematics

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Programmes involving Mathematics

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Second Examination

Final Examination

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Final Examination

2MVA 06 25667 Level I

LI Multivariable & Vector Analysis

2MVA3 06 35172 Level H

LH Multivariable & Vector Analysis

January Examinations 2022-23

Three Hours

Full marks will be obtained with complete answers to all FOUR questions. Each question carries equal weight. You are advised to initially spend no more than 45 minutes on each question and then to return to any incomplete questions if you have time at the end. An indication of the number of marks allocated to parts of questions is shown in square brackets.

No calculator is permitted in this examination.

Section A

1. (a) Consider the following function at point $\mathbf{a} = (1, 2, 3)$,

$$f(x, y, z) = x^3 y^2 z.$$

- (i) Determine a formula for the directional derivative for f at $\mathbf{a}$ along the direction of a unit vector $\mathbf{u}$.
- (ii) Calculate the direction and rate of steepest ascent for f at $\mathbf{a}$.
- (iii) Calculate the direction and rate of steepest descent for f at $\mathbf{a}$.

[8]

- (b) Find and classify all the critical points of the following function:

$$f(x, y) = (y - 2)x^2 - y^2.$$

[8]

- (c) In the partial differential equation,

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = 0,$$

change the variables from (x, y) to (u, v) , where

$$u = \tan^{-1}(y/x), \quad v = \ln \sqrt{x^2 + y^2},$$

and $(x, y) \neq (0, 0)$.

[9]

2. (a) Find the equation of the plane which passes through the point $(-2, 1, 7)$ and is perpendicular to vector $\vec{a} = (1, -8, 5)$. [4]
- (b) Consider a circle of radius $R = 1$ centered at the origin in the (x, y) -plane. A particle moves along the circle at a constant speed. Prove, showing all steps, that the particle's velocity is always orthogonal to its acceleration. [7]
- (c) Consider $\vec{r} = (x, y, z)$, where $(x, y, z) \neq (0, 0, 0)$. Calculate, showing all steps, $\text{div}\left(\frac{\vec{r}}{r}\right)$, where $r = |\vec{r}|$. [8]
- (d) Consider the vector field $\vec{F}(x, y, z) = (x(z - y), y(x - z), z(y - x))$. Find the outward flux of $\vec{F}$ through a unit sphere centered at the point $P = (1, -1, 2)$. [6]

Section B

3. (a) Consider the double integral

$$I = \iint_D \frac{1}{(x^2 + y^2)^{3/2}} dx dy,$$

where the domain D is bounded by $x = 1, x = 2, y = 0$ and $y = \sqrt{3}x$.

- (i) Express the integral as repeated integrals using Cartesian coordinates.
- (ii) Express the integral as repeated integrals using polar coordinates.
- (iii) Calculate the double integral you have written using one of the two coordinate systems.

[10]

- (b) Consider the following triple integral

$$J = \iiint_S e^{(x^2 + y^2 + z^2)^{3/2}} dx dy dz,$$

where the domain S is bounded by the following two surfaces

$$z = \sqrt{3(x^2 + y^2)}, \quad z = \sqrt{4 - (x^2 + y^2)}.$$

- (i) Express the integral as repeated integrals using cylindrical coordinates.
- (ii) Express the integral as repeated integrals using spherical coordinates.
- (iii) Calculate the triple integral you have written using one of the two coordinate systems.
- (iv) Using the result of (iii), or otherwise, calculate the following integral

$$K = \iiint_S \left(e^{(x^2 + y^2 + z^2)^{3/2}} + \sin(x) \cos(y) \right) dx dy dz.$$

- (v) Using the result of (iii), or otherwise, calculate the following integral

$$L = \iiint_T e^{(x^2 + y^2 + z^2)^{3/2}} dx dy dz,$$

where the domain T is defined as

$$T = \{(x, y, z) : (x, y, z) \in S \text{ and } y \geq 0\}.$$

[15]

4. (a) Consider points $P_1 = (1, 1)$, and $P_2 = (2, 6)$ in the (x, y) -plane. Let C_1 be a straight line between the points P_1 and P_2 and C_2 be an arc of the parabola $y = 2x^2 - x$ between the points P_1 and P_2 . Find the difference $I_2 - I_1$ between the following integrals [12]

$$I_1 = \int_{C_1} (x+y)^2 dx - (x-y)^2 dy, \quad \text{and} \quad I_2 = \int_{C_2} (x+y)^2 dx - (x-y)^2 dy.$$

- (b) Demonstrate, showing all steps, that the vector field $\vec{F}(x, y, z) = (2x + yz, xz, xy + 2z)$ is a gradient field. Hence, find a potential function $f(x, y, z)$ for $\vec{F}$ by evaluating line integrals. [13]

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LI/LH Multivariable & Vector Analysis

Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats and outer-wear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) **MUST** be placed in the designated area.
- Check that you **DO NOT** have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches **MUST** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are **NOT** permitted to use a mobile phone as a clock. If you have difficulty in seeing a clock, please alert an Invigilator.
- You are **NOT** permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately.
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to the Student Conduct procedures.